Lesson notes · for you

How good is my model?

Data science · Level 3 · Lesson 4

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- See why accuracy can mislead.
- Fill a confusion matrix.
- Choose a threshold.

Activity: A 2 × 2 table

1. Warm-up (5 min): The forecast says 30% rain. Say whether you would take an umbrella, and why.
2. Count (10 min): With a partner, make up ten days, each with a rain forecast score from 0 to 1 and whether it rained, then fill a confusion matrix at a threshold of 0.5.
3. Decide (10 min): Try a lower threshold, count again and choose one for a school trip.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which mistake is worse for a trip?
- Why can accuracy mislead?
- What happens when the threshold falls?

Check yourself

- Name the two kinds of mistake and give a school example of each.

Try it this week
Keep a simple rain forecast for a week and note whether it rained each day. Fill a 2 × 2 table and decide which mistake was worse for you. Rarely rains where you live? Forecast something that happens on some days, like a power cut or the bus being late.

Worksheet

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Name _____ Class _____ Date _____

1 **What happens when we lower the threshold for a warning?**

◯ Nothing changes ◯ More are caught, with more false alarms ◯ Fewer pupils are caught

2 **Match each outcome to its name.**

Flagged, and really positive	• • • •	False negative
Flagged, but negative	• • • •	True positive
Not flagged, but positive	• • • •	False positive

3 **Count TP, FP and FN, then print precision and recall, each rounded with round(x, 2).**

```
real = [1, 0, 1, 1, 0, 0, 1, 0]
pred = [1, 1, 0, 1, 0, 1, 1, 0]
tp = fp = fn = 0
print(tp, fp, fn)
```

Write your program:

4 **With imbalanced classes, high accuracy guarantees a useful model.**

◯ True ◯ False

5 **A test screens for a serious illness. Which matters most?**

◯ High recall, with a low threshold ◯ High precision, with a high threshold ◯ Accuracy only

✎ **Think of an app that warns you about something. Which mistake would annoy you more: a false alarm or a miss?**

Check your answers

How good is my model?

1 **More are caught, with more false alarms**

A lower bar warns more pupils, rightly and wrongly.

2 **Flagged, and really positive — True positive**
Flagged, but negative — False positive
Not flagged, but positive — False negative

The matrix crosses prediction with truth.

3

```
real = [1, 0, 1, 1, 0, 0, 1, 0]
pred = [1, 1, 0, 1, 0, 1, 1, 0]
tp = fp = fn = 0
for r, p in zip(real, pred):
    if r and p: tp += 1
    if p and not r: fp += 1
    if r and not p: fn += 1
prec = round(tp / (tp + fp), 2)
rec = round(tp / (tp + fn), 2)
print(prec, rec)
```

TP = 3, FP = 2, FN = 1: precision 3 ÷ 5 = 0.6 and recall 3 ÷ 4 = 0.75.

4 **False**

Predicting the majority class every time can score high and catch nothing.

5 **High recall, with a low threshold**

Missing a sick person costs more than a false alarm that a second test can clear.